EXP 6:

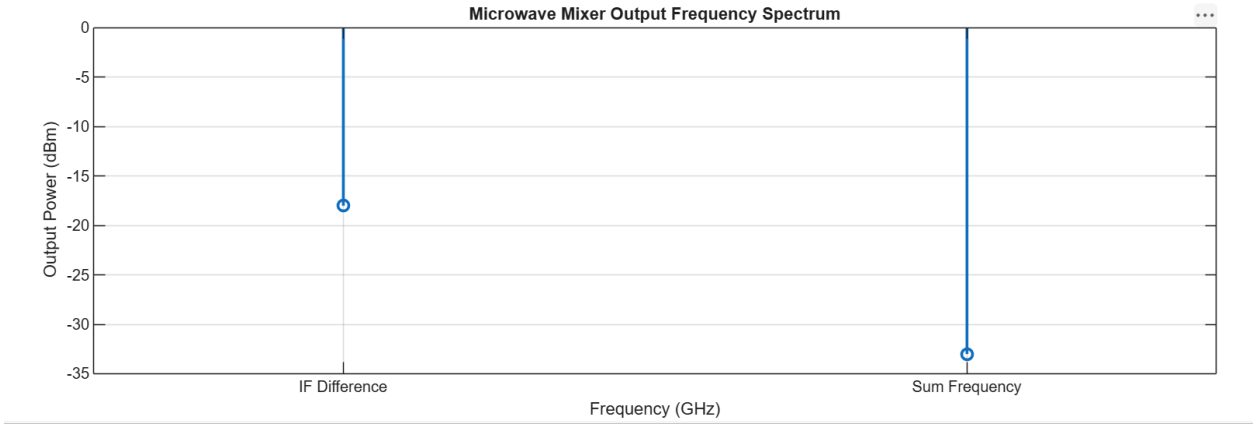
Simulation-Based Investigation of Microwave Semiconductor Devices and Frequency-Conversion Characteristics of a Microwave Mixer

CODE:

```
clc;
clear;
close all;
% ECA13 - Microwave Engineering
% Microwave Mixer Frequency Conversion Analysis
% INPUT PARAMETERS
fRF = 10;          % RF frequency in GHz
fLO = 9.5;         % LO frequency in GHz
PRF = -10;         % RF input power in dBm
PLO = 7;           % LO power in dBm
ZIF = 50;          % IF load impedance in ohms
% Frequency Conversion
fIF = abs(fRF-fLO);
fsum = fRF+fLO;
% Simulated mixer conversion loss
% Assumed passive mixer conversion loss
CL = 8;            % Conversion loss in dB
% IF output power
PIF = PRF-CL;
% Simulated conversion products
P_RF = PRF;
P_LO = PLO;
P_IF = PIF;
P_SUM = PIF-15;    % Illustrative sum-frequency suppression
% Dominant conversion product
f = [fIF, fsum];
P = [P_IF, P_SUM];
[~,idx] = max(P);
dominant_frequency = f(idx);
% COMMAND WINDOW - SIMULATED VALUES ONLY
fprintf('\nMICROWAVE MIXER SIMULATION RESULTS\n');
fprintf('=====================================\n');
fprintf('RF Frequency            = %.2f GHz\n',fRF);
fprintf('LO Frequency            = %.2f GHz\n',fLO);
fprintf('IF Difference Frequency  = %.2f GHz\n',fIF);
```

```
fprintf('Sum Frequency          = %.2f GHz\n',fsum);
fprintf('IF Output Power         = %.2f dBm\n',PIF);
fprintf('Conversion Loss         = %.2f dB\n',CL);
fprintf('IF Load Impedance       = %.2f ohms\n',ZIF);
fprintf('Dominant Conversion Product = %.2f GHz IF\n',...
  dominant_frequency);
fprintf('=====================================\n');
% FREQUENCY SPECTRUM
figure('Color','w');
stem(f,P,'LineWidth',1.8,'MarkerSize',7);
xlabel('Frequency (GHz)');
ylabel('Output Power (dBm)');
title('Microwave Mixer Output Frequency Spectrum');
grid on;
xticks([fIF fsum]);
xticklabels({'IF Difference','Sum Frequency'});
```

OUTPUT:



VERIFICATION:

Exp-6

Given

RF freq $f_{RF} = 10GHz$

LO frequency $f_{LO} = 9.5GHz$

RF input power $P_{RF} = -10dBm$

LO power $P_{LO} = 7dBm$

IF load impedance $Z_{IF} = 50\Omega$

Diff freq $f_{IF} = |f_{RF} - f_{LO}|$

$= |10 - 9.5| = 0.5GHz$

Sum freq $f_{sum} = f_{RF} + f_{LO}$

$= 10 + 9.5 = 19.5GHz$

Conversion loss

$CL = P_{RF,in} - P_{IF,out}$

$8 = -10 - P_{IF,out}$

$\boxed{P_{IF,out} = -18dBm}$

Percentage freq deviation

$$\% \text{ deviation} = \frac{|\text{Practical} - \text{Theoritical}|}{\text{Theoritical}} \times 100$$

$$= \frac{|0.5 - 0.5|}{0.5} = 100 = 0\%$$

Conversion loss deviation

$$= \frac{|8 - 8|}{8} \times 100 = 0\%$$